

Education

Ph.D. in Computing Science

(Jan 2024 - Current)

Simon Fraser University (SFU)

Supervisor: Andrea Tagliasacchi

Courses: 3D Vision, Computer Vision

Ph.D. in Engineering Science

(Sep 2021 - Apr 2023)

Simon Fraser University (SFU)

Supervisor: Ivan V. Bajić

Courses: Information Theory, Deep Learning Systems, Stochastic Systems,
Linear System Theory, Multimedia Communications, Digital Image Processing

M.Sc. in Electrical Engineering

(Sep 2017 - Sep 2020)

University of Tehran

Thesis: Control of Computer's Mouse Using Hand Gesture Recognition

Supervisors: Ahmad Kalhor, Samad Sheikhaei

B.Sc. in Electrical Engineering (Control)

(Sep 2012 - Sep 2017)

Amirkabir University of Technology (Tehran Polytechnic)

Thesis: Driver's Consciousness Level Analysis Using EEG Signals

Supervisor: Mohammad A. Khosravi

Online Courses

Coursera

Deep Neural Networks with PyTorch, Neural Networks and Deep Learning, Improving Deep Neural Networks
Structuring Machine Learning Projects, Convolutional Neural Networks, Introduction to Calculus
Convolutional Neural Networks in TensorFlow, Introduction to TensorFlow, Sequence Models, Machine Learning

Research Experience

Research Assistant at GrUVi Lab

(Jan 2024 - Current)

Simon Fraser University

Developing robust 3D reconstruction model for indoor touring using casual capture from 360° dual-fisheye camera, addressing challenges from distractors and occlusions (Ongoing).

Designed a NeRF-guided training pipeline for 3D Gaussian Splatting, eliminating the need for SfM initialization and improving reconstruction fidelity in complex and dynamic environments (Completed).

Under Supervision of Andrea Tagliasacchi

Focused Expertise: Novel View Synthesis

Research Assistant at Multimedia Lab

(Feb 2022 - Apr 2023)

Simon Fraser University

Developed scalable image codecs for human and machines, primarily optimizing the efficiency of base and enhancement layers.

Contributed significantly to the development of four research papers.

Under Supervision of: Ivan V. Bajić

Focused Expertise: Learnable Image Compression, Compressed Vision, Image and Video Coding

Research Assistant at Advanced Circuits for Data Communication Lab

(Feb 2018 - Sep 2020)

University of Tehran

Developed hand gesture recognition application and collected a hand dataset containing four gestures.

Under Supervision of: Ahmad Kalhor & Samad Sheikhaei

Focused Expertise: Hand Gesture Recognition, Object Detection, Image Classification

Publications

Evaluating Alternatives to SfM Point Cloud Initialization for Gaussian Splatting

Yalda Foroutan, Daniel Rebain, Kwang Moo Yi, Andrea Tagliasacchi

Under Submission. 📄

Base Layer Efficiency in Scalable Human-Machine Coding

Yalda Foroutan, Alon Harell, Anderson Andrade, Ivan V. Bajic

IEEE International Conference on Image Processing, ICIP 2023, Kuala Lumpur, Malaysia. 📄

Rate-Distortion Theory in Coding for Machines and its Application

Alon Harell, Yalda Foroutan, Nilesch A. Ahuja, Parual Datta, Bhavya Kanzariya, V. Srinivasa Somayazulu, Omesh Tickoo, Anderson Andrade, Ivan V. Bajic

Submitted in *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. 

VVC+M: Plug and Play Scalable Image Coding for Humans and Machines

Alon Harell, Yalda Foroutan, Ivan V. Bajic

IEEE International Conference on Multimedia and Expo Workshops, ICMEW Workshops 2023, Brisbane, Australia. 

Conditional and Residual Methods in Scalable Coding for Humans and Machines

Anderson Andrade, Alon Harell, Yalda Foroutan, Ivan V. Bajic

IEEE International Conference on Multimedia and Expo Workshops, ICMEW Workshops 2023, Brisbane, Australia. 

Control of Computer Pointer using Hand Gesture Recognition in Motion Pictures

Yalda Foroutan, Ahmad Kalhor, Saeid Mohammadi Nejati, Samad Sheikhaei

arXiv 2020. 

Academic Projects

3D Vision (CMPT895)

SFU

Course Mini-projects

Moving Least Squares (MLS)

3D Morphable Face Model; Rigid Shape Matching, Rigid Iterative Closest Point (ICP), and Non-Rigid Fitting

Neural Field Reconstruction

Computer Vision (CMPT762)

SFU

Course Mini-projects

Digit Recognition with Convolutional Neural Networks

MNIST Digit Recognition with Convolutional Neural Networks

Optimizing CIFAR-100 Image Classification

Detection, Semantic Segmentation, Instance Segmentation

3D Reconstruction

Deep Learning (ENSC813)

SFU

Course Final Project

Finding the Minimum Bitrates for a Computer Vision Task

Multimedia Communication (ENSC424)

SFU

Course Final Project

Evaluating the Efficiency and Limitations of CANF-VC: Enhancing Performance for Shot Transitions, Motion Information, and Reference Frames

Information Theory (ENSC808)

SFU

Seminar

Presentation and Proving Equations of Information Bottleneck Paper

Digital Image Processing (ENSC895)

SFU

Course Final Project

Segmentation of Breast Cancer Imaging

Neural Networks and Deep Learning

University of Tehran

Course Mini-projects

Design of a CNN Classifier for Fashion MNIST Dataset

Implementation of RNN for Stock Market Prediction

Text Generation Based on Shakespeare's Book with RNN Networks

DCGAN, WGAN, and ACGAN Implementations on Fashion MNIST and CIFAR-10 Datasets

Other Python Projects

Independent Projects

Face Filters: Changing Eye Color, Eye Size, and Putting 3D Sunglasses

Enhancing Low-Light Images and Increasing Image Resolution using GANs

Low light Image Enhancement using GANs

Increasing Image Resolution

Skills

Programming Languages	Python, MATLAB, C/C++, Julia
Frameworks and Libraries	PyTorch, TensorFlow, OpenCV
Other Tools	Bash, HTML/CSS, R

Teaching Experience

2025	Computer Vision (CMPT762)	SFU
2024	Introduction to Engineering Analysis (ENSC180)	SFU
2023	Capstone B (ENSC440)	SFU
2023	Graphical Communication (ENSC204)	SFU
2023	Capstone A (ENSC405)	SFU
2023	Computational Neuroscience (Certificate)	Neuromatch Academy
2023	Capstone A (ENSC405)	SFU
2022	Graphical Communication (ENSC204)	SFU
2021	Computational Neuroscience (Certificate)	Neuromatch Academy
2021	Deep Learning (Certificate)	Neuromatch Academy
2020	Neural Networks and Deep Learning	University of Tehran
2020	Electronic 1	University of Tehran
2019	Electronic 1	University of Tehran
2019	Electronic 3	University of Tehran

Awards and Honors

2025	CMPT Graduate Fellowships from Computing Science department (CMPT) at SFU	Canada
2025	Graduate Fellowship from SFU	Canada
2025	PhD Research Scholarship from SFU	Canada
2024	CMPT Graduate Fellowships from Computing Science department (CMPT) at SFU	Canada
2024	Graduate Fellowship from SFU	Canada
2024	PhD Research Scholarship from SFU	Canada
2022	Talent Bursary from Alberta Machine Intelligence Institute (amii)	Canada
2020	Ph.D. Admission from University of Tehran	Iran
2008	2nd Place in Iran Mathematics Olympiad	Iran
2006	1st Place in Abadan Mathematics Olympiad	Iran
2005	Exceptional Talent Recognition by National Organization for Development of Exceptional Talents (NODET)	Iran

The electronic version contains hyperlinks.

For the extended version and more information, feel free to visit <https://youlenda.github.io>